

Vaibhavi Shambhu Shetty

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EDUCATION

North Carolina State University, Raleigh, NC

Aug 2023 - May 2025

Masters, Computer Science

GPA: 3.87/4.00

Relevant Coursework: Cloud Computing Architecture, Parallel Systems, Software Engineering, Algorithms, ML with Graphs, Database Management Systems, Operating Systems, Object-Oriented Design and Development

Vishwakarma Institute of Technology, Pune, India

Aug 2019 - May 2023

Bachelor of Technology, Information Technology (*Honors in AI and Data Analytics*)

CGPA: 9.68/10.00

TECHNICAL SKILLS

Languages: Python, Java, C++, C#, JavaScript, TypeScript, SQL, Ruby, Bash/Shell Scripting, HTML/CSS

Cloud & DevOps: Microsoft Azure DevOps, AWS (EC2, S3), Docker, Kubernetes, Git, Jenkins, Ansible, Terraform

Framework: ReactJS, Node.js, Flask, TensorFlow, PyTorch, REST APIs, Spring Boot, Bootstrap, Flutter

Databases: MySQL, Firebase, PostgreSQL, MongoDB, Relational Databases, NoSQL, Redis

PUBLICATIONS

- Stylized NFT Progressive Neural Paintings using Brush Stroke Prediction – [IEEE \(2023\)](#)
- Non-Facial Video Spatiotemporal Forensic Analysis Using Deep Learning Techniques – [PETL Taiwan \(2022\)](#)

WORK EXPERIENCE

Graduate Research Assistant, NC State University, NC

Aug 2024 - May 2025

- Processed 100K+ genomic records by designing ETL pipelines in Python, boosting ML accuracy to 90%+.
- Streamlined data ingestion & transformation workflows, cutting preprocessing time by 25%.
- Built ReactJS dashboards improving data accessibility for 50+ experiments, while mentoring 8 undergraduates.

Junior Associate Technology, Synechron Inc, FL

May 2024 - Aug 2024

- Built anomaly detection pipelines (Isolation Forest) on 1M+ records, using AWS S3 + EC2 for storage and computing, reducing false positives by 20%
- Automated validation workflows in Python, improving compliance and reliability.
- Integrated DevOps + MLflow for deployment automation and monitoring reducing model release time by 40%.

Software Developer, Appficiency, Pune, India

Jan 2023 - Jun 2023

- Developed and deployed a new scalable microservices ERP application feature with ReactJS, Node.js, and Docker using automated CI/CD pipelines ensuring 99.9% uptime.
- Optimized backend throughput by 40% via async APIs, caching, and DB optimizations, cutting query latency from 250ms → 150ms.
- Engineered robust, secure REST APIs with a strong focus on data privacy, compliance, and reliability.

ACADEMIC PROJECTS

Course Manager – Python, MySQL, ReactJS, TypeScript

- Built a dynamic React + Redux interface with Flask backend, delivering a responsive and scalable platform.
- Engineered a normalized MySQL database, improving query efficiency by 30%.

Expertiza (Open Source) – ReactJS, TypeScript, Ruby on Rails

- Developed role-based UIs and real-time notifications using React + Rails for 500+ active users.
- Built secure access control with RSpec-tested backend APIs, reducing authorization errors by 40%.

High-Performance Canny Edge Detection – C++, CUDA, MPI, OpenMP

- Parallelized image processing pipeline with MPI + OpenMP, reducing execution time by 65%.
- Achieved 5× GPU speedup with optimized CUDA kernels for Gaussian filtering and hysteresis stages.

CERTIFICATIONS

- Google Cloud Certified – 30 Days of Google Cloud Program
- AWS Certified Developer – Associate (In Progress)